

Question number	Answer	Mark
4(a)(i)	The only correct answer is C <i>A is incorrect because cellulose is made of beta glucose</i> <i>B is incorrect because cellulose is made of beta glucose and has 1 - 4 glycosidic bonds only</i> <i>D is incorrect because cellulose has 1 - 4 glycosidic bonds only</i>	(1)

Question number	Answer	Mark
4(a)(ii)	The only correct answer is A <i>B is incorrect because bacteria do not have nuclei</i> <i>C is incorrect because bacteria do not have nuclei</i> <i>D is incorrect because enzymes are released by exocytosis</i>	(1)

Question number	Answer	Additional guidance	Mark
4(b)(i)	because different leaves have different water content (1)	ACCEPT water content fluctuates no {energy / organic matter / nutritional content} in water biomass is the organic matter biomass does not include water	(1)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	• {62 / 63 / 64} and 8 (1) • 0.60 / 0.61 / 0.62 (1)	Bald answer of 0.60 / 0.61 / 0.62 = 2 marks Bald answer of 0.6 / 0.611111 / 0.622222 = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
4(b)(iii)	A description that includes two of the following points: • both decompose at the same (approximate) rate in the first {30 / 60} days (1) • decomposition levels off after 30 days in the white oak but continues to decrease in the sugar maple (1) • sugar maple leaves decompose faster (overall) (1)	ACCEPT converse for white oak leaves refs to {changes in dry mass / breakdown} for decomposition ACCEPT sugar maple leaves decompose more	(2)

Question number	Answer	Additional guidance	Mark
4(c)(iii)	An answer that includes the following points: • {sugar / nutrient} was exhausted / leaves produce toxins (1) • the bacteria were dying (faster than reproducing) (1) OR • competition for {nutrients / space} between microorganisms (1) • bacteria were dying (faster than reproducing) (1) OR • bacteria are eaten (1) • by organisms living on the white oak leaves / predators (1) OR • the value at 90 days is an anomaly (1) • and therefore indicates a decrease that did not happen (1)	ACCEPT insufficient nutrients IGNORE food ACCEPT bacteria were in the {death / decline} phase IGNORE cannot reproduce ACCEPT microorganisms producing toxins ACCEPT bacteria were in the {death / decline} phase	(2)

